Visual Inertial Navigation System for Autonomous Robotic Surveillance

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BACKGROUND

This research addresses the need for precise and repeatable autonomous navigation for robotic surveillance. Industrial applications require robots to follow specific patrol routes consistently, but GPS can be unreliable indoors or near large structures. Our approach is to develop a robust Visual Inertial Navigation (VIN) system that does not depend on continuous GPS. The system first utilizes Visual-Inertial SLAM (Simultaneous Localization and Mapping) to map a desired route. This map is then converted into a series of waypoints, enabling the robot to autonomously navigate and re-trace the exact path repeatedly. This poster details the successful validation of our system's accuracy and repeatability within a motion capture testbed.

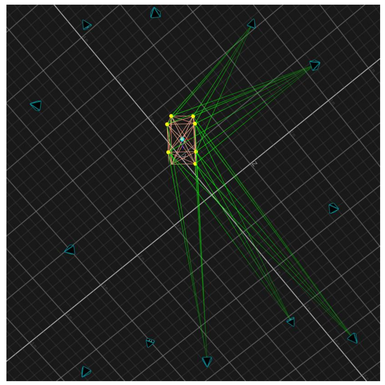


Figure 1: MoCap Tracking Markers

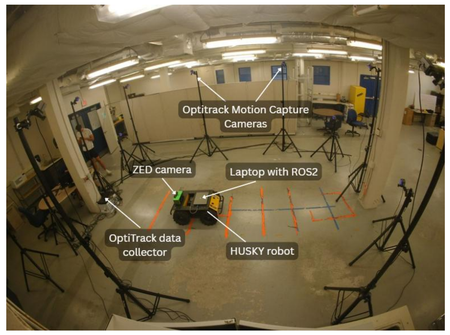


Figure 2: Lab Testbed

SYSTEM DESIGN

Our system architecture is built upon a Clearpath Husky A200 mobile base, equipped with a ZED X stereo camera for perception. An onboard computer runs ROS 2 Humble on Ubuntu 22.04, managing all software components. The core of our approach is a Visual Inertial Navigation (VIN) pipeline that processes synchronized image and IMU data from the camera. We leverage the official zed_ros2_wrapper and the robust pose estimation node within the ZED SDK to compute the robot's real-time pose (position and orientation).

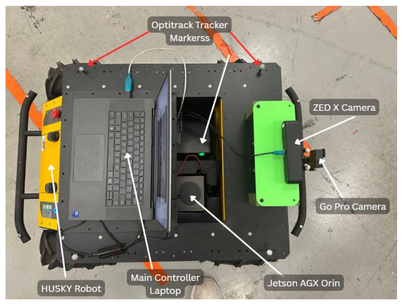


Figure 3: Husky Setup

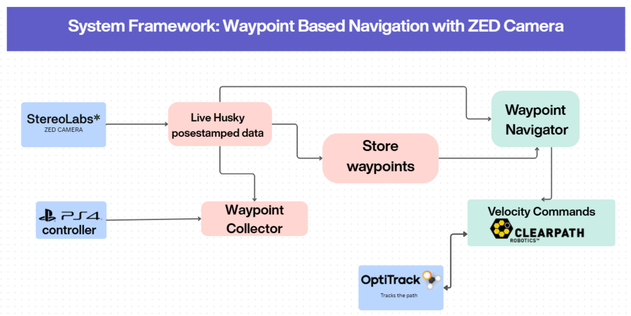


Figure 4: System Framework Diagram

RESULTS

Our Visual Inertial Navigation (VIN) system demonstrated high-fidelity performance and successful path completion in the motion capture testbed. The following results validate the accuracy of the system's localization and the effectiveness of the PD controller.

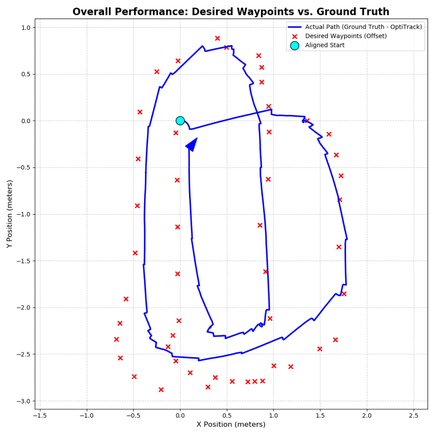


Figure 7: Overall performance

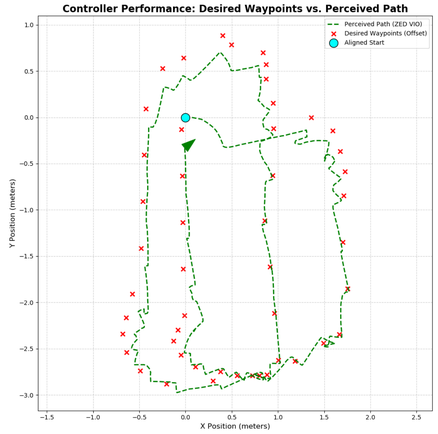


Figure 8: Controller Performance



Figure 9: Localization Performance

To quantify the system's accuracy, the Absolute Trajectory Error (ATE) was calculated between the VIN-estimated path and the OptiTrack ground truth.

- Root Mean Square Error (RMSE): [X.XX] meters
- Mean Error: [X.XX] meters
- Standard Deviation: [X.XX] meters

These low error metrics confirm the system's ability to maintain a precise and reliable pose estimate, making it a strong foundation for repeatable autonomous patrols.

EXPERIMENTAL VALIDATION

To validate our system, the robot was programmed to autonomously follow a predefined series of waypoints within the motion capture arena. The robot's VIN-estimated trajectory and the ground-truth trajectory from the OptiTrack system were recorded simultaneously for direct comparison

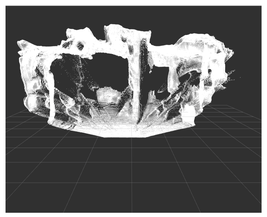


Figure 5: ZED camera Depth Perception

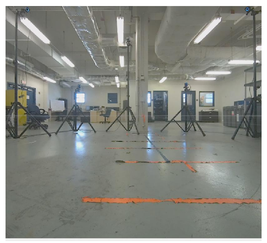


Figure 6: Zed Camera RGB View

CONCLUSION

We successfully developed and validated a robust Visual Intertidal Navigation (VIN) system, integrating a ZED X camera on a Husky robot using ROS2. Rigorous testing against a motion capture system confirmed the system's high-fidelity tracking and minimal odometry drift. This work establishes a reliable foundation for autonomous surveillance in challenging environments like power stations. Future efforts will focus on implementing more sophisticated control strategies to enhance path-tracking performance.

ACKNOWLEGEMENTS

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